

01

Proofs

Combine quantifier rules with the proof steps we already know. Find the right proof type for your problem.

$\forall$ $\exists$

EXERCISE 1

Write Complete Proof using Rules of Inference.

I eat spinach (S) or ice cream (I).

If I study logic (L), then you pass the exam (E).

If I eat ice cream, then I study logic.

If I eat spinach, then I play golf (G).

You fail the exam.

o1 therefore I play golf.



RETRIEVAL

So far

Lecture 6: valid arguments, named rules, and justified proofs.

$$p \rightarrow$$
$$q, p \vdash$$
$$q$$

Valid arguments

No row makes every premise true
and the conclusion false.

**MP, MT,
HS, DS**

Implication & OR rules

Modus ponens, modus tollens,
syllogisms, resolution.

$$p \wedge$$
$$q \vdash p$$

AND rules

Simplification and conjunction
split and combine facts.

Lecture 7 assumes fluency in each of these rules.

$$\therefore$$

Building & auditing proofs

Justify every line; catch the
consequent and antecedent
errors.

THE ROUTE

Today

Extend inference to quantifiers,
then start choosing a proof strategy.

01

APPLY

Quantified rules

Instantiate, generalize, and introduce a fresh witness.

$$\forall, \exists$$

02

EVALUATE

Contraposition

A parity proof shows when the reversed route is easier.

$$P \rightarrow Q \equiv \neg Q \rightarrow \neg P$$

03

CREATE

Direct proof

Build from definitions: divisibility, primes, and composites.

$$p \rightarrow q$$

QUANTIFIED INFERENCE

Universal instantiation

Universal = for all. Instantiation = taking a particular instance.

$$\forall x P(x)$$

c belongs to the domain

$$\therefore P(c)$$

KNOW

Every lab robot is registered.

CHOOSE

Alice is a lab robot.

CONCLUDE

Alice is registered.

A universal claim applies to each allowed object.

INstantiate the whole formula

The registered robot

LINE	STATEMENT	REASON
1	$\forall x (\text{Bot}(x) \rightarrow \text{Reg}(x))$	Premise
2	$\text{Bot}(\text{Alice})$	Premise
3	$\text{Bot}(\text{Alice}) \rightarrow \text{Reg}(\text{Alice})$	1: universal instantiation
4	$\text{Reg}(\text{Alice})$	2, 3: modus ponens

The domain is all objects. Bot identifies the robots.

QUANTIFIED INFERENCE

Universal generalization

Universal = for all. Generalization = forming a general claim.

$P(c)$ proved

c arbitrary

$\therefore \forall x P(x)$

ARBITRARY INTEGER

Let c be any integer.

PROVE

$c + 0 = c$.

GENERALIZE

Every integer x satisfies $x + 0 = x$.

No undischarged assumption may single out c .

QUANTIFIED INFERENCE

Existential generalization

Existential = there exists. Generalization = forming a general claim.

$$P(c)$$

c belongs to the domain

$$\therefore \exists x P(x)$$

A WITNESS

Alice solved the maze.

CONCLUSION

Someone solved the maze.

MEANING

At least one is enough.

A known instance provides an existential witness.

EXERCISE 2

S : submitted. T : tested. E : eligible.

Domain: students. $\forall x ((S(x) \wedge T(x)) \rightarrow E(x)), S(\text{Alice}), T(\text{Alice})$.

01 Apply Universal Instantiation:

02 Apply Conjunction:

03 Apply Modus Ponens:

QUANTIFIED INFERENCE

Existential instantiation

Existential = there exists. Instantiation = introducing a particular instance.

$$\exists x P(x)$$

Choose fresh c : $P(c)$

EXISTENCE

Some robot solved the maze.

A TEMPORARY NAME

Call a qualifying robot c .

RESTRICTION

c is not a preselected robot.

Keep witness-dependent reasoning within its scope.

EXERCISE 3

True or False: one line justification!

Domain: people. $P(x)$: plays piano. $Q(x)$: codes.

01 $\exists x P(x), \exists x Q(x) \quad \therefore \exists x (P(x) \wedge Q(x))?$

02 $\exists x (P(x) \wedge Q(x)) \quad \therefore \exists x P(x)?$

LINE	STATEMENT	REASON
1	$\forall x (R(x) \rightarrow C(x))$	Premise
2	$\forall x (C(x) \rightarrow A(x))$	Premise
3	$\exists x R(x)$	Premise
4	$R(c)$	3: existential instantiation
5	$R(c) \rightarrow C(c)$	1: universal instantiation
6	$C(c)$	4, 5: modus ponens
7	$C(c) \rightarrow A(c)$	2: universal instantiation
8	$A(c)$	6, 7: modus ponens
9	$R(c) \rightarrow A(c)$	5, 7: hypothetical syllogism
10	$\forall x A(x)$	8: universal generalization

Given the first three premises, audit every remaining line.
Line 10 is illegal; lines 1-9 are all correct.

02

Proof by contraposition

The same conditional can offer a different route.

$$\begin{aligned} &P \rightarrow Q \\ \equiv &\neg Q \rightarrow \neg P \end{aligned}$$

DEFINITIONS FOR THE NEXT PROOF

Even and odd integers

Even

$$n = 2k$$

For some $k \in \mathbb{Z}$. An integer multiple of two.

Odd

$$n = 2k + 1$$

For some $k \in \mathbb{Z}$. One more than an even integer.

Every integer is exactly one of even or odd.

EXERCISE 4

A parity claim

For $n \in \mathbb{Z}$, prove: if $3n + 2$ is odd, then n is odd.

01 What is the contrapositive?

02 Which definition starts that proof?

CONTRAPOSITION

The parity proof

STEP	STATEMENT	REASON
1	Let $n \in \mathbb{Z}$ be arbitrary.	Set the domain
2	Assume n is even: $n = 2k, k \in \mathbb{Z}$.	Negate the original conclusion
3	$3n + 2 = 3(2k) + 2$	Substitution
4	$= 2(3k + 1)$	Algebra
5	$3k + 1 \in \mathbb{Z}$, so $3n + 2$ is even.	Definition of even
6	Thus $3n + 2$ odd implies n odd.	Contraposition

Only n even was assumed for the contrapositive.

EXERCISE 5

A new parity proof

For $n \in \mathbb{Z}$, prove: if $5n + 4$ is odd, then n is odd.

01 State the contrapositive.

02 Complete the proof.

PROOF METHODS

What counts as a proof?

A valid argument establishing a mathematical statement.

Axiom / postulate

Accepted foundation.

Hypothesis / premise

Assumed for this argument.

Theorem

Established by a proof.



A conjecture is a claim still awaiting proof.

PROOF METHODS

Divides means “is a multiple of”

Here $m, n \in \mathbb{Z}$ and $m \neq 0$.

$$m \mid n$$

$$\Leftrightarrow n = km \text{ for some } k \in \mathbb{Z}$$



$$3 \mid 6 \text{ because } 6 = 2 \cdot 3.$$

WORKED EXAMPLE • TRANSITIVITY

A multiple of a multiple

Given $a \mid b$ and $b \mid c$, prove $a \mid c$.

Unpack

$$b = ra, c = sb$$

$$r, s \in \mathbb{Z}$$

Substitute

$$c = s(ra) = (sr)a$$

Finish

$$sr \in \mathbb{Z} \Rightarrow a \mid c$$



EXERCISE 6

Keep the common divisor

Let $d \mid a$ and $a \mid b$. For any $x, y \in \mathbb{Z}$, prove:

$$d \mid (xa + yb)$$



PROOF METHODS

Prime or composite?

For an integer $n > 1$:

Prime

Positive divisors: 1 and n

Example: 7

Composite

A divisor m with $1 < m < n$

Example: $12 = 3 \cdot 4$

1 is neither prime nor composite.

WORKED EXAMPLE

One factor settles 289

So 289 is composite.

$$289 = 17 \cdot 17$$



One proper divisor is enough.